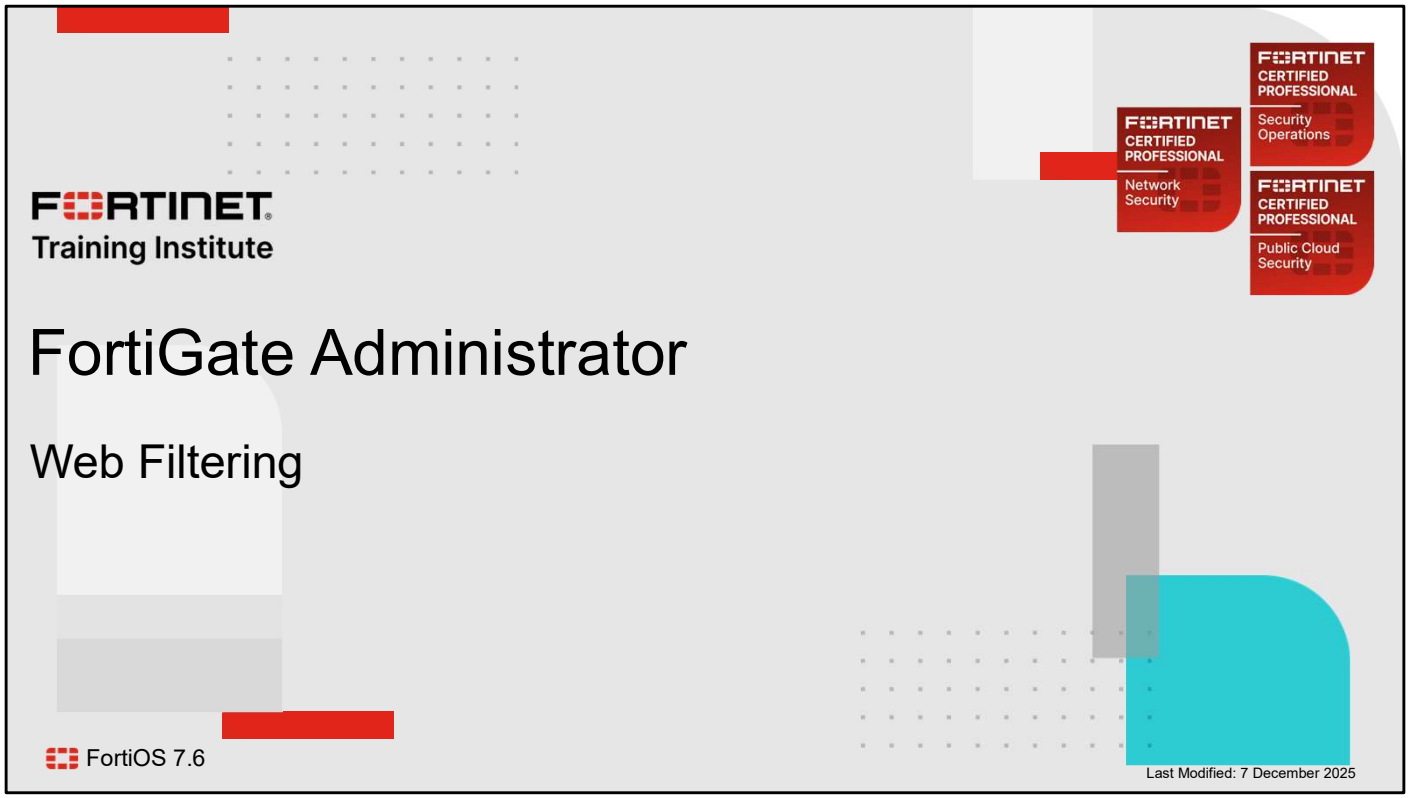

In this lesson, you will learn how to configure web filtering on FortiGate to control web traffic in your network.

Lesson Overview

- Inspection Modes
- Web Filtering Basics
- Troubleshooting

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In this lesson, you will learn about the topics shown on this slide.

Inspection Modes

Objectives

- Select the correct inspection mode flow or proxy based on security needs

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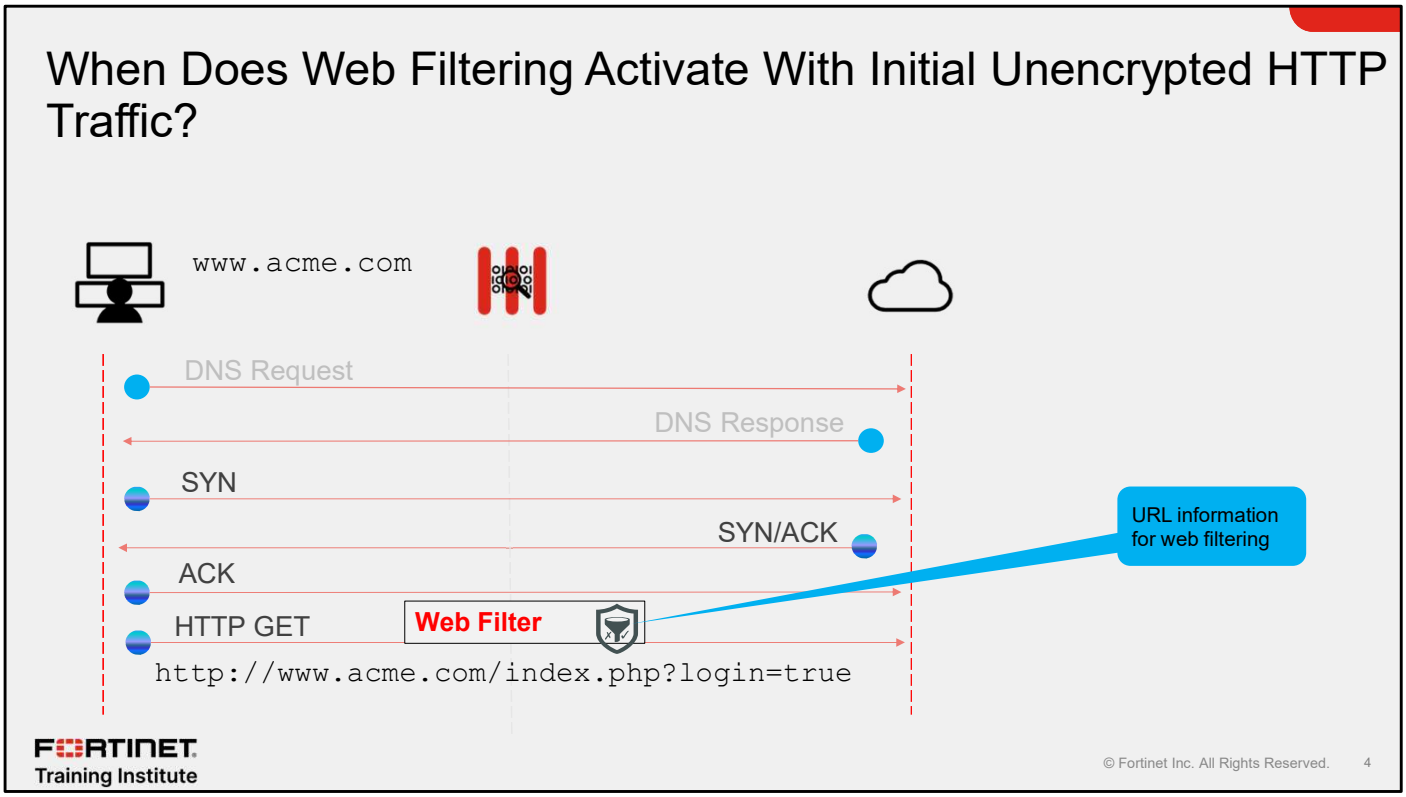
After completing this lesson, you should be able to achieve the objective shown on this slide.

By demonstrating competence in understanding inspection modes, you will be able to implement the appropriate inspection modes to support the desired security profiles.



FortiGate web filtering helps stop infections from malware sites and prevent communication if an infection occurs.

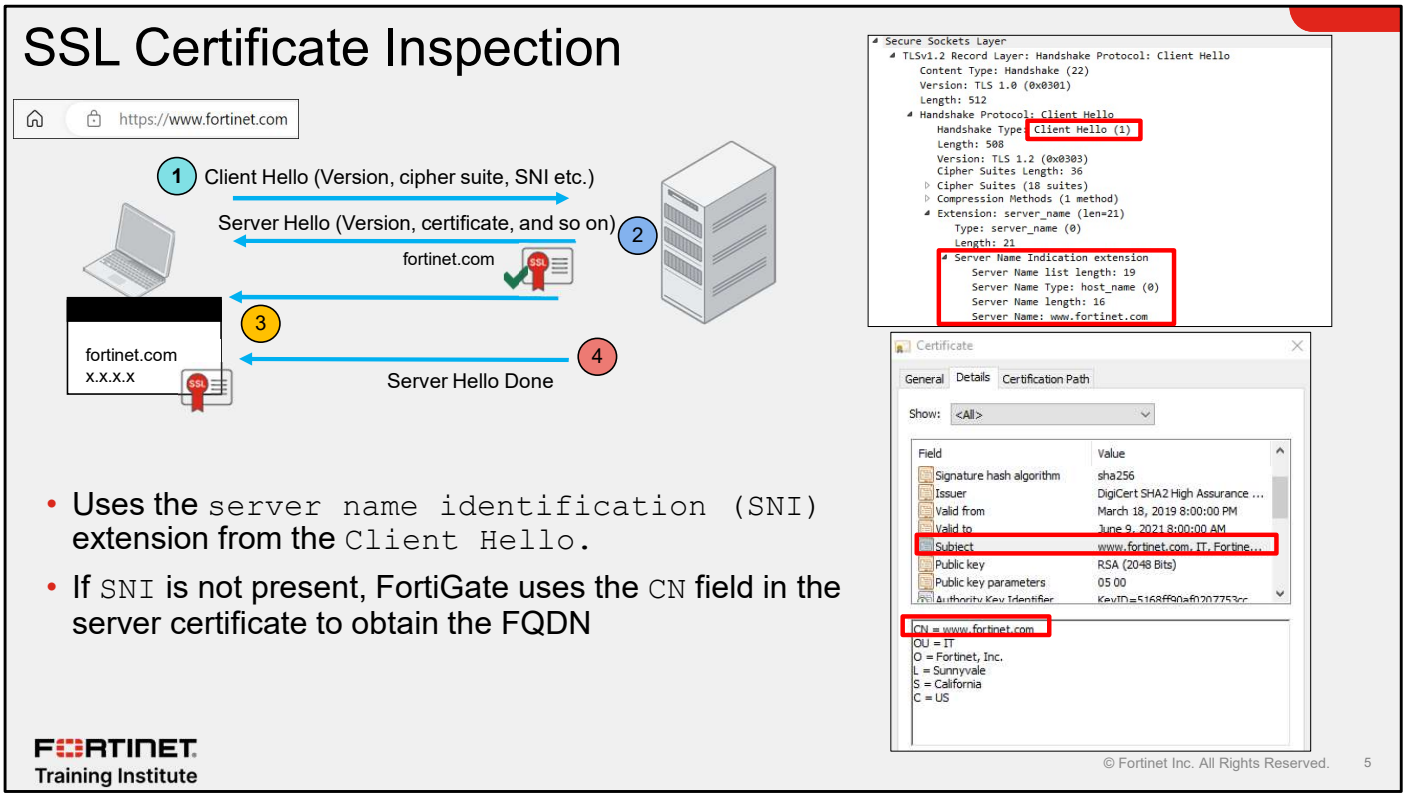
After a TCP connection is established, the user requests the content of a specific web site (for example, www.acme.com) using an HTTP GET request.

FortiGate checks the HTTP GET request to collect the information shown in the slide and sends a real-time request to the FortiGuard servers to determine the category of the website. Based on the category information from the FortiGuard Distribution Network (FDN), FortiGate performs the specific action based on the profile applied to the policy.

In FortiOS, there are three main components of web filtering:

- Web content filtering: blocks web pages containing words or patterns that you specify.
- URL filtering: uses URLs and URL patterns to block or exempt web pages from specific sources, or block malicious URLs discovered by FortiSandbox.
- FortiGuard Web Filtering service: provides many additional categories you can use to filter web traffic.

If you filter by domain, sometimes the filter blocks too much. For example, each blog on `tumblr.com` is viewed as a separate piece of content because they have different authors. So, when filtering blogs, you might want to be more specific and block blogs based on specific parts of the URL, for example, `tumblr.com/hacking`.



The stages of the TLS handshake process are as follows:

1. The "Client Hello" message, which includes SNI, is sent to the server.
2. The server receives the packet from the client and replies with the "Server Hello" message, which may initially include empty information in the SNI response.
3. The server sends the authentication details required by the SNI request.
4. Finally, the server sends a "Server Hello Done" message.

For encrypted protocols, FortiGate requires additional inspection. When using SSL certificate inspection, FortiGate inspects the headers up to only the SSL/TLS layer. FortiGate doesn't decrypt or inspect any encrypted traffic. Using this method, FortiGate inspects only the initial unencrypted SSL handshake.

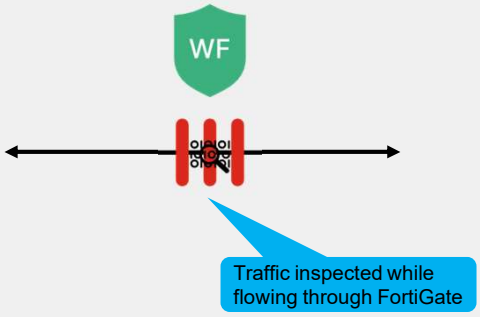
If the SNI field exists, FortiGate uses it to obtain the FQDN to rate the site. If the SNI isn't present, FortiGate retrieves the FQDN from the CN field of the server certificate.

In some cases, the CN server name might not match the requested FQDN. For example, the value in the common name (CN) field in the digital certificate of `youtube.com` is `google.com`. So, if you connect to `youtube.com` from a browser that doesn't support SNI, and FortiGate uses the SSL certificate inspection method, FortiGate assumes, incorrectly, that you are connecting to `google.com`, and uses the `google.com` category instead of the category for `youtube.com`.

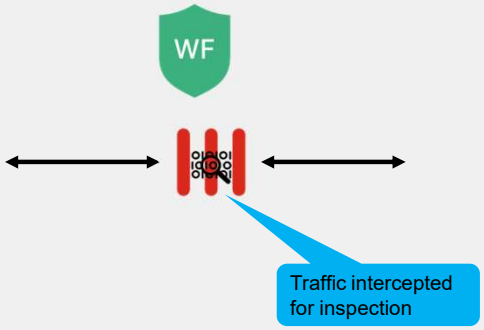
SSL certificate inspection works correctly with web filtering, because the full payload does not need to be inspected.

Web Filtering Inspection Modes

- Flow-based inspection
  - Default inspection mode
  - Requires fewer processing resources
  - Faster scanning



- Proxy-based inspection
  - More thorough inspection
  - Provides additional options
  - More resource intensive



- Firewall policy must include an SSL inspection profile and a web filter profile

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You can configure web filtering in flow-based or proxy-based inspection mode.
The flow-based inspection method examines the file as it passes through FortiGate, without buffering it. Packets are analyzed and forwarded as they are received, without waiting for the complete file or web page.
The advantages of flow-based inspection mode are:

- The user sees a faster response time for HTTP requests compared to proxy-based inspection mode.
- There is less chance of a time-out error caused by the server at the other end responding slowly.
- From FortiOS 7.4.4 and later flow-based web filter profiling supports safe search and restrictions on YouTube and Vimeo access.

The disadvantages of flow-based inspection mode are:

- A number of security features that are available in proxy-based inspection mode are not available in flow-based inspection mode.
- Fewer actions are available based on the categorization of the website by FortiGuard services.

On the other hand, proxy-based scanning refers to transparent proxy. It's called transparent because, at the IP layer, FortiGate is not the destination address, but FortiGate *does* intercept the traffic. When proxy-based inspection is enabled, FortiGate buffers traffic and examines it *as a whole*, before determining an action. Because FortiGate examines the data as a whole, it can examine more points of data than it does when using flow-based inspection.
The proxy analyzes the headers and may change the headers, such as HTTP host and URL, for web filtering. If a security profile decides to block the connection, the proxy can send a replacement message to the client. This adds latency to the overall transmission speed.
Make sure that both an SSL inspection profile (such as the certificate-inspection profile) and a web filter profile are selected in the associated firewall policy.

Lesson Progress

- ✓ Inspection Modes
- Web Filtering Basics
- Troubleshooting

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Good job! You now understand inspection modes.

Now, you will learn about web filtering basics.

Web Filtering Basics

Objectives

- Configure certificate inspection for web filtering
- Configure a web filter profile in flow-based inspection mode
- Configure a web filter profile in proxy-based inspection mode
- Configure FortiGuard categories
- Configure a URL filter

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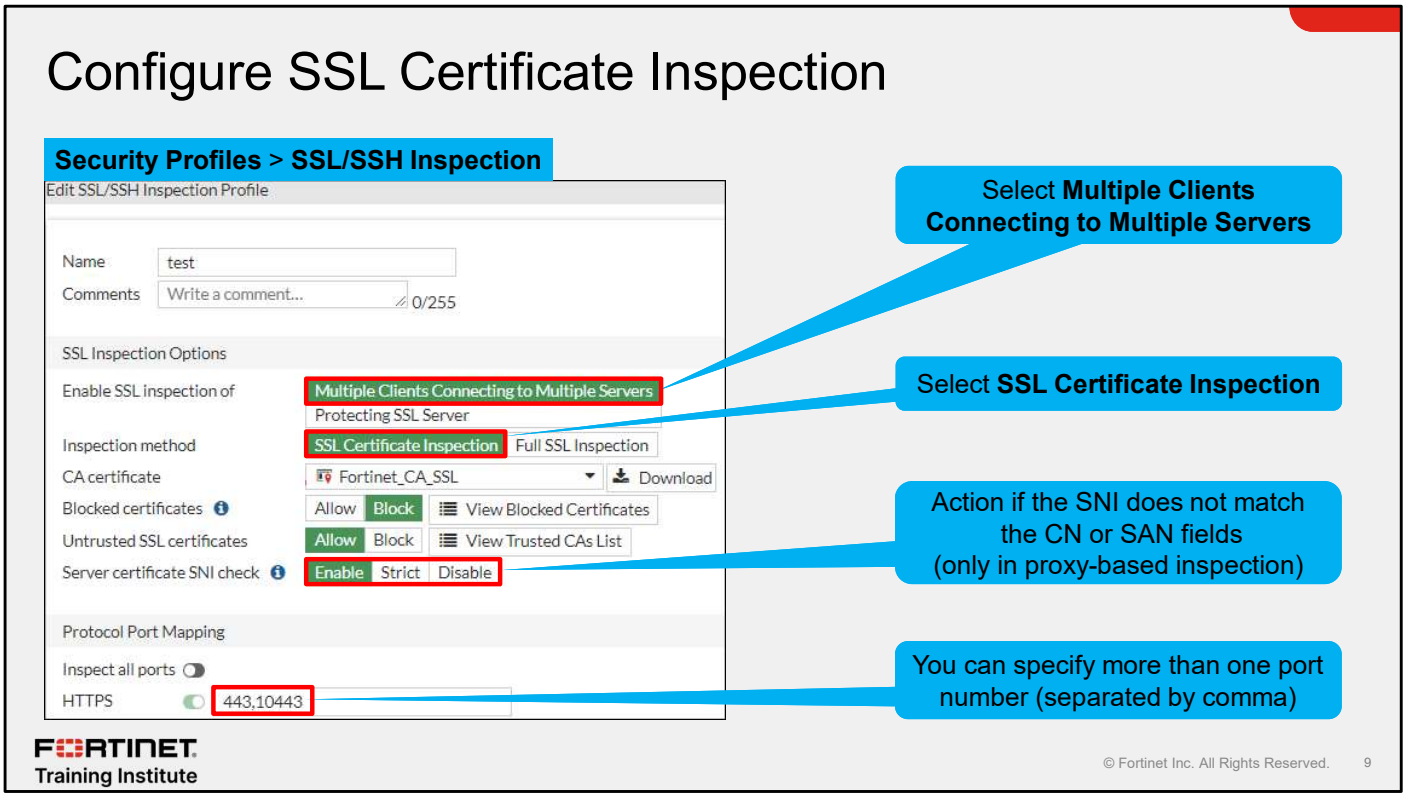
After completing this lesson, you should be able to achieve the objectives shown on this slide.

By demonstrating competence in web filtering configuration, you will be able to implement the web filter profile in an effective manner.

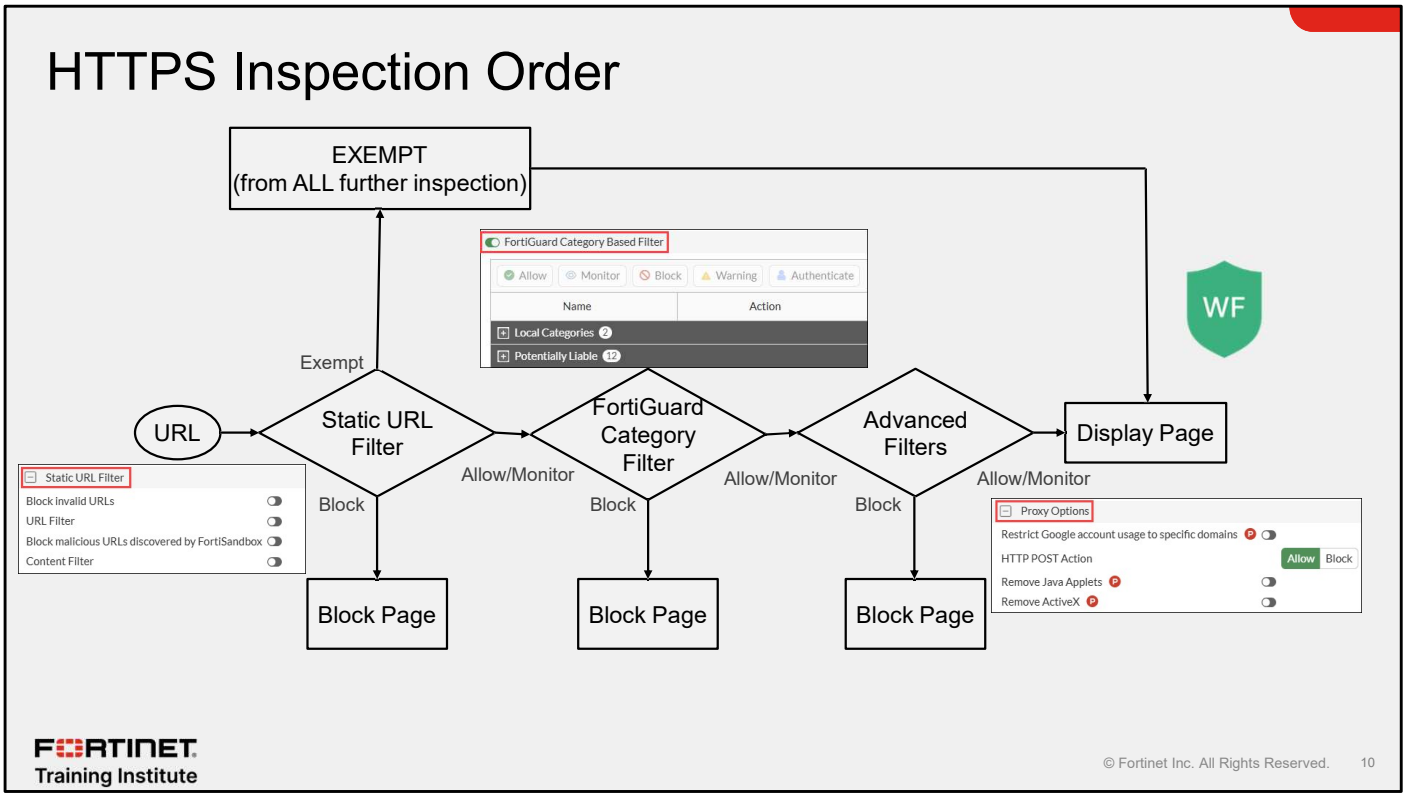


FortiGate has a read-only preconfigured profile for SSL certificate inspection named **certificate-inspection**. If you want to enable SSL certificate inspection, select this profile when configuring a firewall policy.

Alternatively, you can create your own profile for SSL certificate inspection by following these steps:

1. On the FortiGate GUI, click **Security Profiles**, and then click **SSL/SSH Inspection**.
2. Click **Create New** to create a new SSL/SSH inspection profile.
3. Select **Multiple Clients Connecting to Multiple Servers**, and then click **SSL Certificate Inspection**.
4. Select the action for **Server certificate SNI check**.

When the **Server certificate SNI check** configuration is **Enable**, FortiGate uses the domain in the CN field instead of the domain in the SNI field if the domain in the SNI field does not match any of the domains listed in the CN and SAN fields. With **Strict**, FortiGate closes the client connection if there is a mismatch. When **SNI check** is **Disable**, FortiGate always rates URLs based on the FQDN.



So, with these different features, what is the inspection order? If you have enabled many of them, the inspection order flows as follows:

1. The local static URL filter
2. FortiGuard category filtering (to determine a rating)
3. Advanced filters (such as safe search or removing Active X components)

For each step, if there is no match, FortiGate moves on to the next check enabled.

Configure Web Filter Profiles—Flow Based

- Apply web filter profile to a flow-based firewall policy

Select **Flow-based**

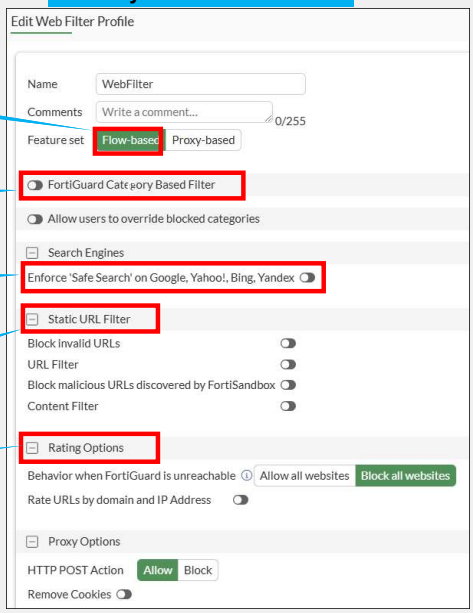
Enable **FortiGuard Category Based Filter** and configure each category

Enable **Safe Search** if needed

Enable and configure **Static URL Filter** if needed

Enable and configure **Rating Options** if needed

Security Profiles > Web Filter



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Now, you will learn more about configuring the web filter profile. You can configure this security profile to use a feature set for proxy-based or flow-based inspection modes. However, depending on the mode you select, the available settings are different. Flow-based inspection has fewer available options.

Selecting rating options allows you to enable two advanced filter features:

- **Allow websites when a rating error occurs**: If you have activated services that require a FortiGuard license (such as FortiGuard filter), but do not have a license, then you will get a rating error message. Use this setting to allow access to websites that return a rating error from the FortiGuard Web Filter service.

- **Rate URLs by domain and IP address**: If you enable this option, in addition to sending only the domain information, FortiGate will always send the URL domain name and the IP address of the TCP/IP packet (except private IP addresses) to FortiGuard for rating. If a different IP address and URL domain category has been returned by the FortiGuard server, then FortiGate uses the rating weight of the IP address or domain name to determine the rating result and decision. This rating weight is hard coded in FortiOS.

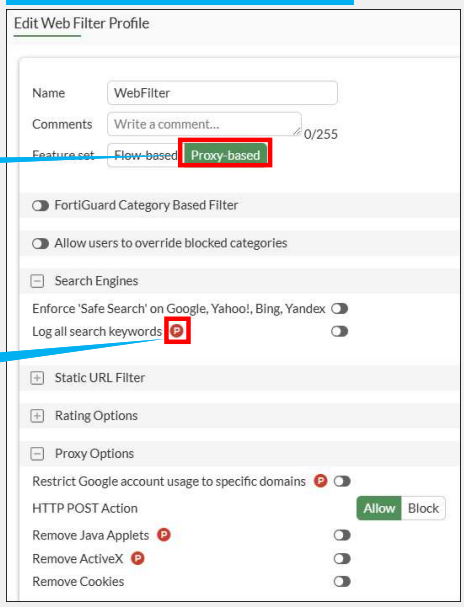
Configure Web Filter Profiles—Proxy Based

- Apply a web filter profile to a proxy-based firewall policy
- Supported on only FortiGate models with more than 2 GB RAM

Select **Proxy-based**

Feature available only in proxy based

Security Profiles > Web Filter



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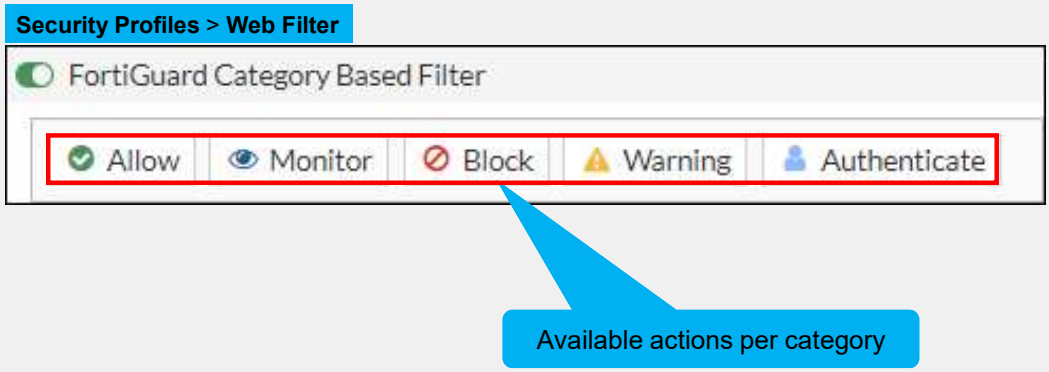
In the example shown on this slide, the security profile is configured to use a proxy-based feature set. It provides features specific to proxy-based configuration.

After you configure your web filter profile, you can apply this profile to the firewall policy configured to use proxy-based inspection mode, so the filtering is applied to your web traffic.

To enhance performance and optimize memory usage, FortiOS no longer supports proxy-related features on FortiGate models with 2 GB of RAM or less.

FortiGuard Category Filter

- Websites split into multiple categories
- Live connection to FortiGuard with active contract required
- Can use FortiManager instead of FortiGuard



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In the web filter profile, Fortiguard category filtering enhances the web filter features. Rather than block or allow websites individually, it looks at the category that a website has been rated with. Then, FortiGate takes action based on that category, not based on the URL.

FortiGuard category filtering is a live service that requires an active contract. The contract validates connections to the FortiGuard network. If the contract expires, there is a two-day grace period during which you can renew the contract before the service ends. If you do not renew, after the two-day grace period, FortiGate reports a rating error for every rating request made. In addition, by default, FortiGate blocks web pages that return a rating error. You can change this behavior by enabling the **Allow websites when a rating error occurs** setting.

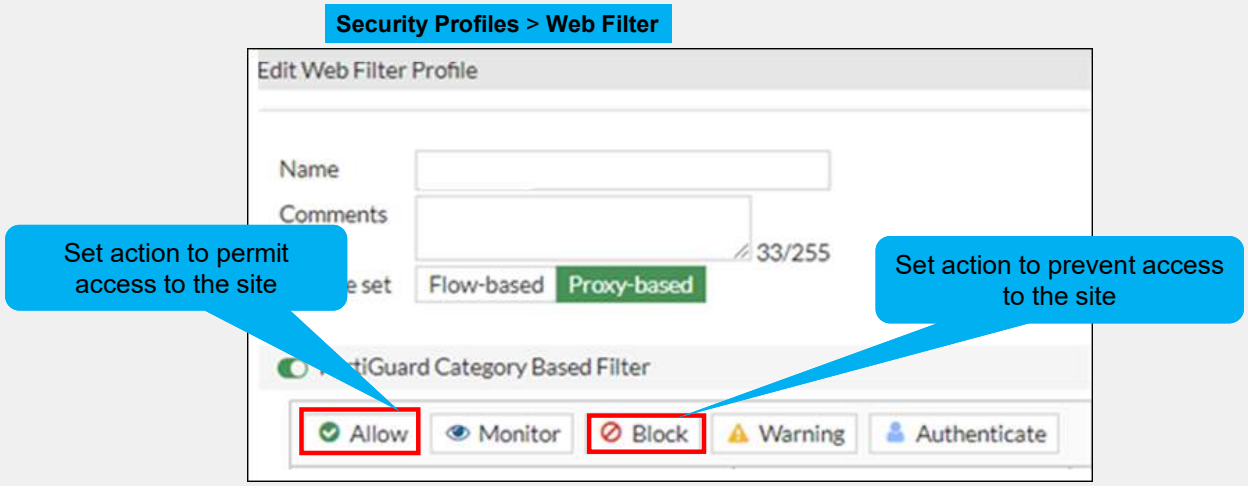
You can configure FortiManager to act as a local FortiGuard server. To do this, you must download the databases to FortiManager, and configure FortiGate to validate the categories against FortiManager, instead of FortiGuard.

You can enable the FortiGuard category filtering on the web filter profile. Categories are listed, and you can customize the actions to perform individually. The actions available are **Allow**, **Monitor**, **Block**, **Warning**, and **Authenticate**.

To review the complete list of categories, visit the FortiGuard web filter website.

Web Filter FortiGuard Category Action—Allow or Block

- Allow or block access web sites



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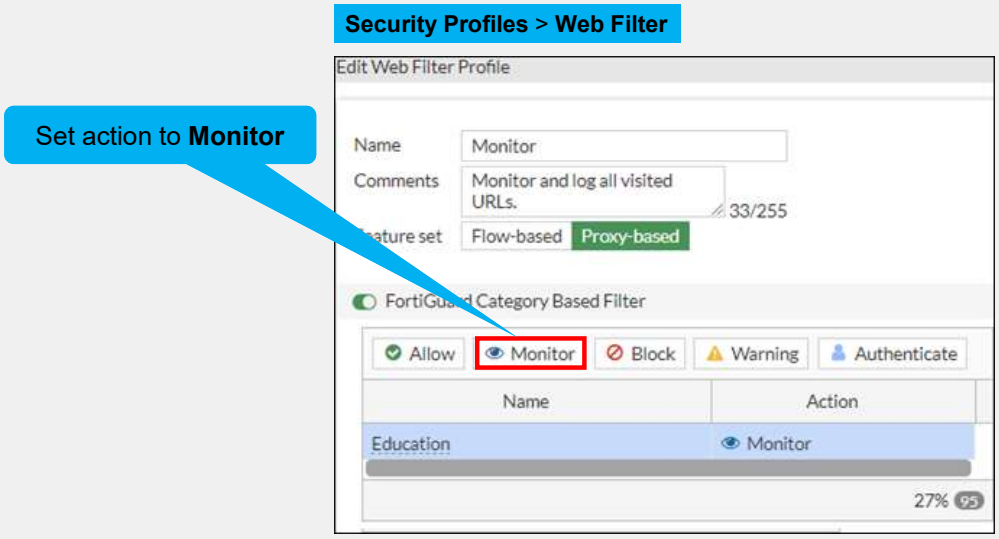
The **Allow** action permits access to the sites in the category but does not create a web filter security log. The **Block** action prevents access to the sites in the category and creates a web filter security log as well. Users trying to access a blocked site will get a message indicating the site is blocked.

Web Filter FortiGuard Category Action—Monitor

- Monitor action allows and logs web sites accesses



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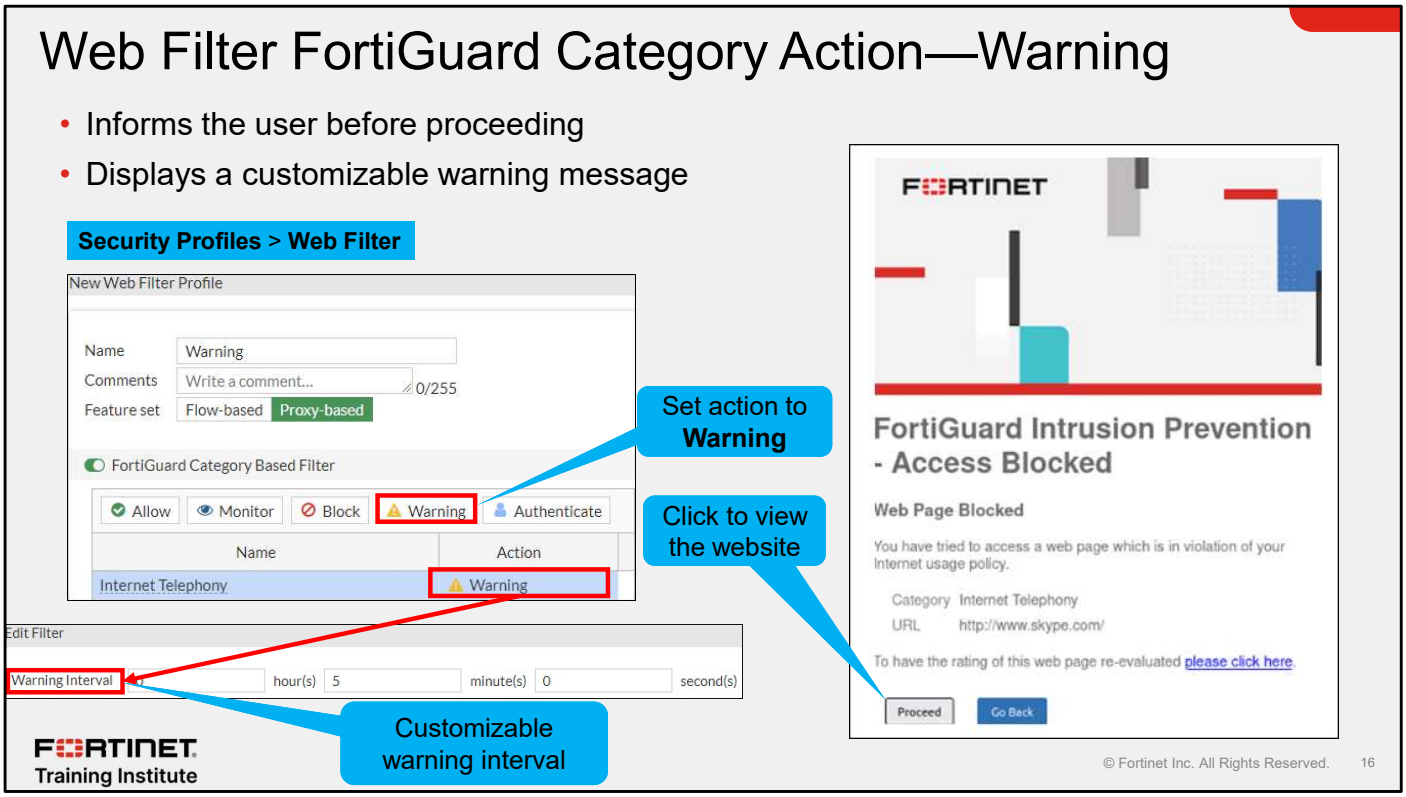
The **Monitor** action allows access to the sites in the category and creates a security log for analysis purposes.



The **Warning** action informs users that the requested website is not allowed by the internet policies. However, the action gives the user the option to proceed to the requested website or return to the previous website.

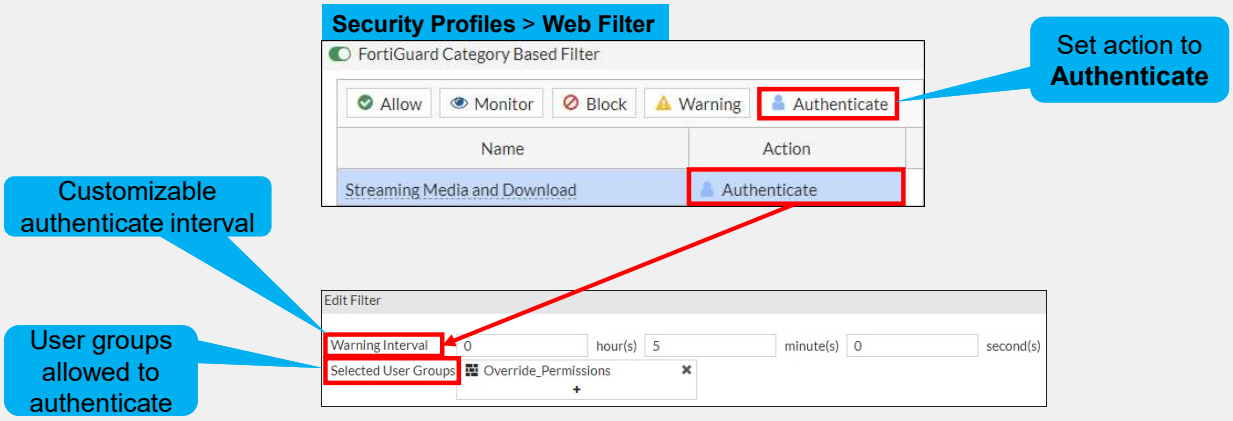
You can customize the warning interval. When the timer expires, FortiGate displays the warning message again, if you access other websites in the same category.

You can customize the warning replacement message. By default, it provides information about the URL and its corresponding category. With this information, the user can click **Proceed** to override the internet usage policy.

The user's first attempt to connect will be blocked, and a corresponding web filter security log will be generated accordingly with action `blocked` and, if the user clicks **Proceed**, another web filter security log will be generated with the action `passthrough` and the message "URL belongs to a category with warnings enabled".

Web Filter FortiGuard Category Action—Authenticate

- To configure the **Authenticate** action:
  - Define users and a group
  - Set action to **Authenticate**
  - Select a user group



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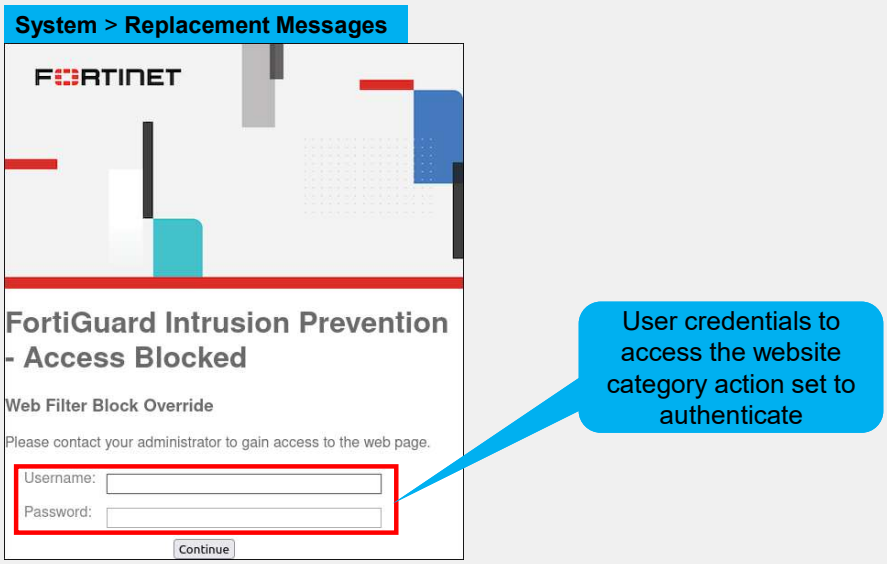
The **Authenticate** action blocks the requested websites, unless the user enters a successful username and password. FortiGate supports local and remote authentication using LDAP, RADIUS, and so on, for web filtering authentication. Choosing this action prompts you to define user groups that are allowed to override the block.

You can also customize the interval of time to allow access. Users are not prompted to authenticate again if they access other websites in the same category until the timer expires.

The user's first attempt to connect will be blocked, and a corresponding web filter security log will be generated. If the user proceeds and successfully authenticates, then another web filter security log will be generated with the action `passthrough` and the message "URL belongs to a category with warnings enabled".

Web Filter FortiGuard Category Action—Authenticate (Contd)

- User credentials requested in message



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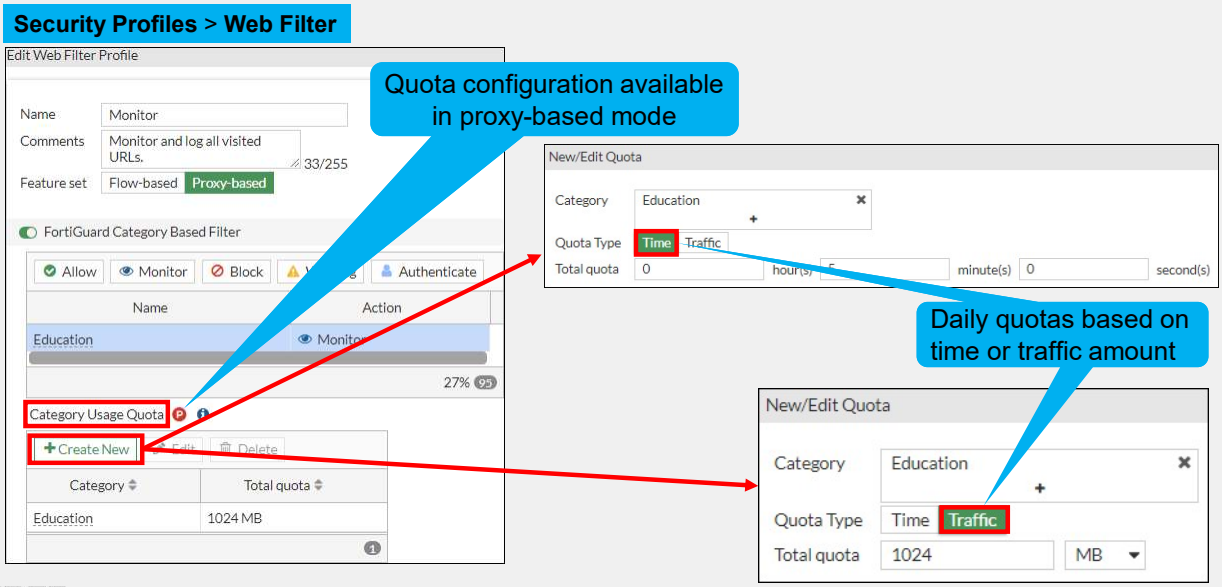
Like the **Warning** action, FortiGate displays a replacement message to proceed and a second one asks for user credentials. You can customize these replacement messages in **System** > **Replacement Messages**.

Web Filter FortiGuard Category Action—Quotas

- Applies to **Monitor**, **Warning** and **Authenticate** actions
- Quotas available only in proxy-based mode



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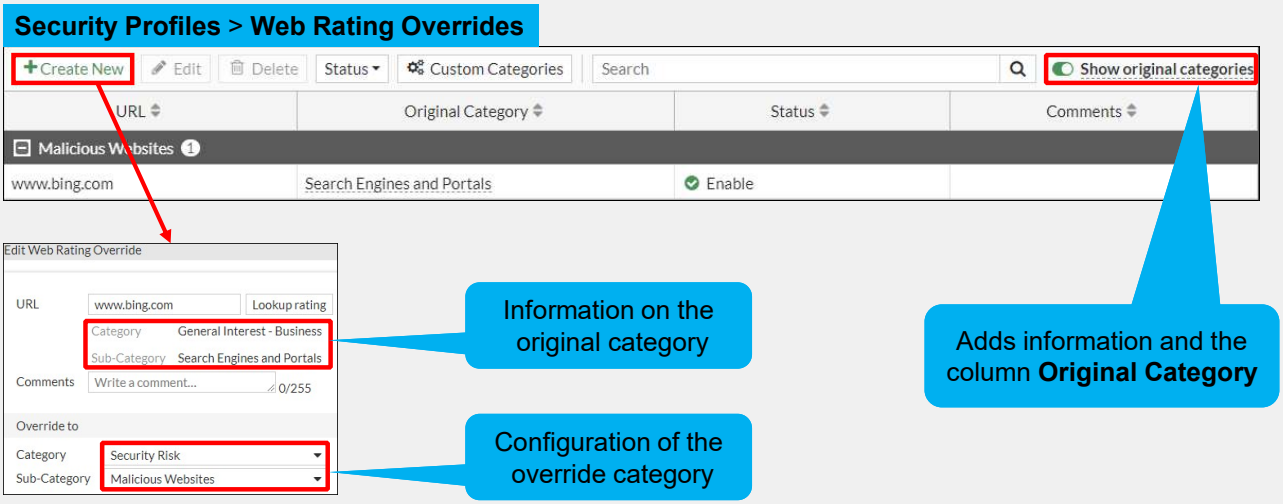
Quotas allow daily access for a specific length of time or bandwidth.
At midnight, quotas reset. Once the daily quota is reached for a category, FortiGate blocks the traffic and displays a replacement message page. You can apply quotas to **Monitor, Warning** and **Authenticate** actions.

Web Rating Override

- Changes a website category, not the category action



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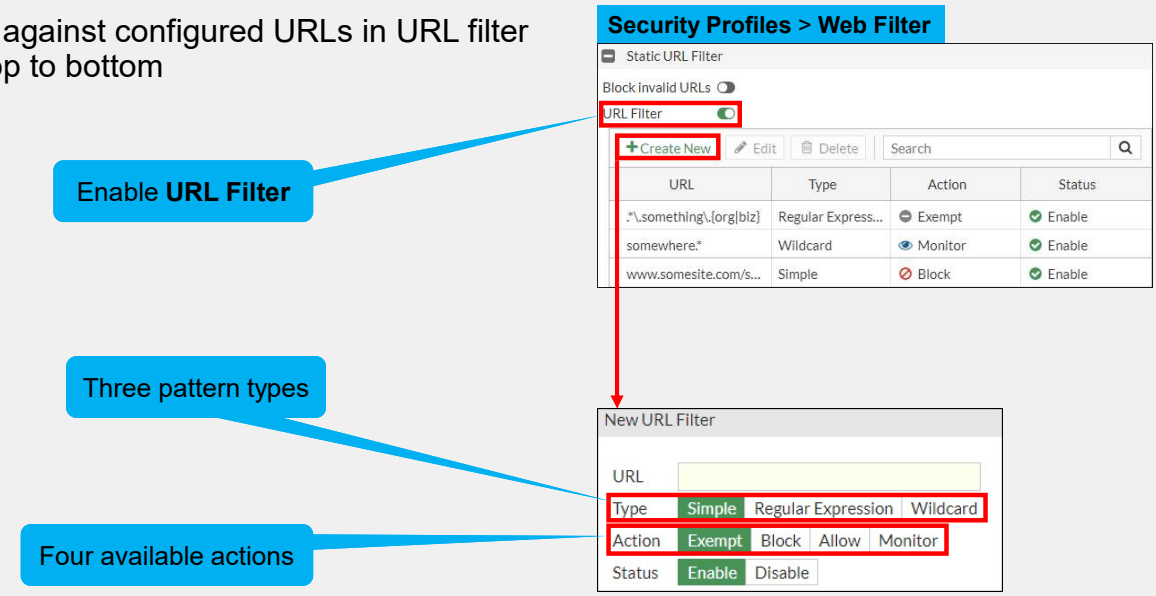
If you consider that a particular URL does not have the correct category, you can ask to re-evaluate the rating in the Fortinet URL Rating Submission website. You can also override a web rating for an exceptional URL in the FortiGate configuration.

Remember that changing categories does not automatically result in a different action for the website. This depends on the settings within the web filter profile.

Configure a URL Filter

- Check against configured URLs in URL filter from top to bottom



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Static URL filtering is another web filter feature, which provides more granularity. Configured URLs in the URL filter are checked from top to bottom against the visited websites. If FortiGate finds a match, it applies the configured action. You can configure one of four actions:

- **Exempt** allows the traffic from trusted sources to bypass all security inspections.
- **Block** denies the attempt and the user receives a replacement message.
- **Allow** permits access. The traffic is passed to the remaining operations, including FortiGuard web filter, web content filter, web script filters, and antivirus scanning.
- **Monitor** allows the traffic while creating log entries. The traffic is still subject to all the other security profile inspections.

To find the exact match, URL filtering has three pattern types: **Simple**, **Regular Expressions**, and **Wildcard**.

Lesson Progress

- ✓ Inspection Modes
- ✓ Web Filtering Basics
- Troubleshooting

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Good job! You now understand the basics of web filtering.

Now, you will learn about troubleshooting.

Troubleshooting

Objectives

- Troubleshoot web filtering issues
- Monitor logs for web filtering events

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After completing this lesson, you should be able to achieve the objectives shown on this slide.

By demonstrating competence in troubleshooting, you will be able to apply troubleshooting techniques to avoid and investigate common issues.

Troubleshooting the FortiGuard Connection

- FortiGuard category filtering requires a live connection

```
FortiGate # diagnose debug rating
Locale       : english

Service      : Web-filter
Status       : Enable
License      : Contract
\
Num. of servers : 1
Protocol        : https
Port            : 8888
Anycast         : Disable
Default servers : Not included

-=- Server List (Wed Sep 20 09:22:42 2023) -=-

IP                      Weight   RTT Flags   TZ   FortiGuard-requests  Curr Lost Total Lost            Updated Time
10.0.1.241                -244     2 I        0                   122          0          0 Wed Sep 20 09:21:55 2023
```

Weight decreases with successful packets

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Category-based filtering requires a live connection to FortiGuard.

You can verify the connection to FortiGuard servers by running the `diagnose debug rating` CLI command. This command displays a list of FortiGuard servers you can connect to, as well as the following information:

- `Weight`: It is based on the difference in time zones between FortiGate and this server to reduce the possibility of using a remote server.
- `RTT`: Return trip time
- `Flags`: `D` (IP returned from DNS), `I` (Contract server contacted), `T` (being timed), `F` (failed)
- `TZ`: Server time zone
- `FortiGuard-requests`: The number of requests sent by FortiGate to FortiGuard
- `Curr Lost`: Current number of consecutive lost FortiGuard requests (in a row, it resets to `0` when one packet succeeds)
- `Total Lost`: Total number of lost FortiGuard requests

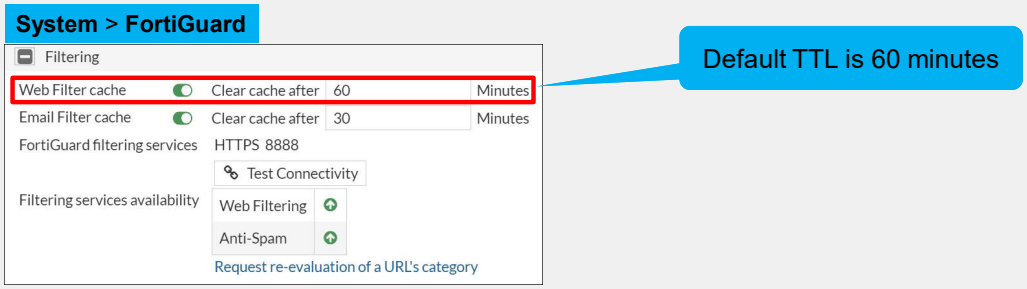
The list is of variable length depending on the FortiGuard Distribution Network and the FortiGate configuration.

Troubleshooting the FortiGuard Connection (Contd)

- Change default FortiGuard or FortiManager communications from HTTPS port 443:
  - Disable FortiGuard anycast setting on CLI to use UDP ports 443, 53, or 8888

```
config system fortiguard
    set fortiguard-anycast {enable|disable}
    set protocol {udp|https}
    set port {8888|53|443}
end
```

- Enable **Web Filter cache** to reduce requests to FortiGuard



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By default, FortiGate is configured to enforce the use of HTTPS port 443 to perform live filtering with FortiGuard or FortiManager. When the `fortiguard-anycast` command is `enable`, the FortiGuard domain name resolves to a single anycast IP address, which is the only entry in the list of FortiGuard servers. By disabling the FortiGuard anycast setting on the CLI, other ports and protocols are available. These ports and protocols query the servers (FortiGuard or FortiManager) on HTTPS port 53 and port 8888, UDP port 443, port 53, and port 8888. If you are using UDP port 53, any kind of inspection reveals that this traffic is not DNS and prevents the service from working. In this case, you can switch to the alternate UDP port 443 or port 8888, or change the protocol to HTTPS, but these ports are not guaranteed to be open in all networks, so you must check beforehand.

If the number of FortiGuard requests is too high, you can also enable **Web Filter cache.** Once enabled, FortiGate maintains a list of recent website rating responses in memory. So, if the URL is already known, FortiGate doesn't send back a rating request. Caching responses reduces the amount of time it takes to establish a rating for a website since the memory lookup is much quicker than packets travelling on the internet.

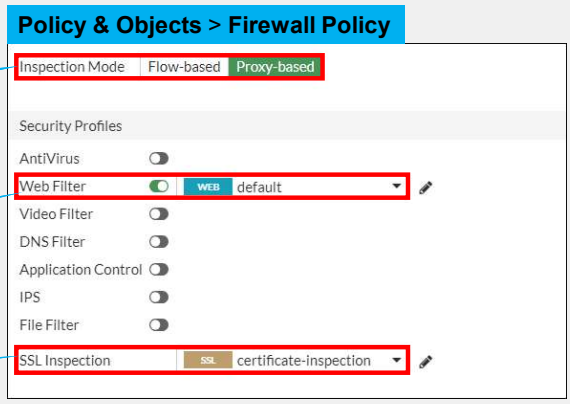
Troubleshooting Web Filtering Issues

- Web filtering not working even with a valid FortiGuard live connection?

Compare **inspection mode** setting with **feature set** in web filter profile

Verify the web filter profile applied

For encrypted protocols, certificate-inspection must be at least selected



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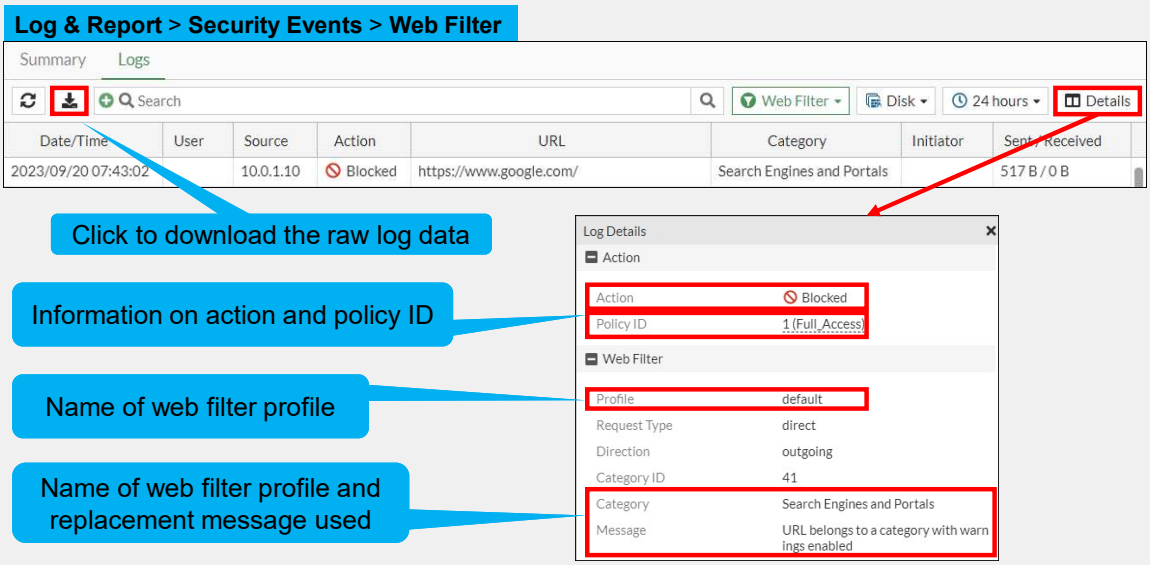
What if you have a live connection to FortiGuard and configured your security profiles, but they are not performing web inspection?

Most of the time, issues are caused by misconfiguration on the device. You can verify them as follows:

- Make sure that the **SSL Inspection** field includes at least one profile with an SSL certification inspection method.
- Make sure that the correct web filter profile is applied on the firewall policy.
- Verify the inspection mode setting with the feature set in the corresponding web filter profile.

Web Filter Log

- Record HTTP traffic activity including action, profile used, category, URL and quota info



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To confirm the correct configuration and web filtering behavior, you can view the web filter logs.

This slide shows an example of a log message. Access details include information about the FortiGuard quota and category (if those are enabled), which web filter profile was used to inspect the traffic, the URL, and more details about the event.

You can also view the raw log data by clicking the download icon. The file downloaded is a plaintext file in a syslog format.

Knowledge Check

1. Which action in the FortiGuard category filter supports the category usage quota?
   - ✓ A. Warning
   - B. Allow

2. Which configuration change is required to use UDP ports 443, 53, or 8888 for the FortiGuard connection?
   - ✓ A. Use the `set fortiguard-anycast disable` command on the CLI.
   - B. Enable the **Override FortiGuard Servers** setting on the GUI.

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Lesson Progress

- ✓ Inspection Modes
- ✓ Web Filtering Basics
- ✓ Troubleshooting

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Congratulations! You have completed this lesson.

Now, you will review the objectives covered in this lesson.

Review

- ✓ Describe FortiOS inspection modes
- ✓ Implement a web filter profile in flow-based and proxy-based inspection modes
- ✓ Work with web filter categories
- ✓ Configure a URL filter for further granularity
- ✓ Troubleshoot common web filtering issues
- ✓ Monitor logs for web filtering events

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This slide shows the objectives that you covered in this lesson.

By mastering the objectives covered in this lesson, you learned how to configure web filtering on FortiGate to control web traffic in your network.